# Lab: Hands That Know the Yarn

## Materials Exploration — Agents

### Objective

Investigate your own hands as active inference agents by comparing how they sense, predict, and respond to different yarn materials. You will develop firsthand awareness of how tactile expertise works and how precision weighting shifts with material familiarity.

### Materials Needed

- 5 yarn samples of different fiber types and weights (e.g., worsted wool, DK cotton, fingering silk, bulky acrylic, textured novelty yarn)
- A crochet hook appropriate for each yarn
- A blindfold or opaque bag
- A notebook for observations
- A timer

### Part 1: The Blindfold Test (20 minutes)

1. Have a partner (or yourself, with eyes closed) place each yarn sample into your hands one at a time.
2. Without looking, handle each yarn for 60 seconds. Record:
3. What fiber do you think it is?
4. How confident are you (1-5 scale)?
5. What specific sensations led to your guess? (Temperature, drag, spring, halo, smoothness, weight)
6. Open your eyes and check the label. How accurate were your predictions?
7. Repeat the exercise, but this time try to assess: yarn weight category, twist tightness, and ply count — still by touch alone.

**Reflection**: Your confidence rating is a direct measure of precision weighting. High confidence means your generative model has strong predictions for that fiber type. Low confidence means your model is uncertain.

## Part 2: The Unfamiliar Material Challenge (20 minutes)

1. Choose the yarn that was most unfamiliar to you from Part 1.
2. Crochet a 15-stitch foundation chain and 5 rows of single crochet with this yarn.
3. While working, pay close attention to:
4. Where do your hands hesitate? Where are they uncertain?
5. How does your tension compare to working with your most familiar yarn?
6. At what point (if ever) do your hands start to feel more comfortable?
7. Now switch to your most familiar yarn and crochet the same swatch. Notice the difference in ease and confidence.

**Reflection**: The contrast between these two experiences reveals the difference between a well-trained generative model (familiar yarn) and a model under construction (unfamiliar yarn).

## Part 3: The Agent-Material Dialogue (15 minutes)

1. Choose two yarns with very different "personalities" — for example, a grippy cotton and a slippery silk.
2. Crochet 3 rows with each, alternating between them.
3. For each yarn, record:
4. How does your grip change?
5. How does your hook speed change?
6. How does your tensioning hand adjust?
7. Does the yarn seem to "cooperate" or "resist"?

**Reflection**: The adjustments your hands make are active inference in action — the agent adapting its actions to minimize prediction error with each material.

## Part 4: Speed and Attention Mapping (10 minutes)

1. Crochet a row with your most familiar yarn while counting your stitches per minute.

2. Crochet a row with the unfamiliar yarn, counting stitches per minute.
3. During each row, rate how much conscious attention your hands require (1 = autopilot, 5 = full concentration).

| Yarn | Stitches/min | Attention (1-5) |
| --- | --- | --- |
| | | |

**Reflection**: Low attention + high speed = low prediction error = low free energy. This is the state that crafters experience as flow.

## Discussion Questions

1. Were you surprised by how much information your hands could extract from yarn through touch alone?
2. What specific sensations were most informative for identifying fiber type? Temperature? Texture? Spring?
3. How did the "dialogue" with different yarns change your hand behavior? Did you notice yourself making adjustments you did not consciously plan?

## Reflection

Write a short paragraph about your hands as agents. What did this lab teach you about the intelligence that lives in your fingertips? How does your experience level affect your hands' ability to predict and respond to different materials?